

EMPIRICAL ARTICLE

Captured Memories: The Impact of First-Person Versus Third-Person Viewpoint Photographs on Remembering Personal Experiences

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People take a staggering number of photos to capture experiences. How do features of the photos people review affect how they will remember these experiences? Across five studies, both in field settings and using respondents' own experiences and photos ($N = 709$), we find that reviewing a single or multiple photos that included the self shifts people's memories toward an observer's memory perspective, rather than that of their original experience. However, when reviewing photos that depict a first-person viewpoint, memories are recalled more from an actor's memory perspective. While prior research explicitly instructed people to adopt a particular viewpoint during recall without photos, or examined the effect of recall involving photos correlationally, we causally demonstrate how this shift can occur naturally, by simply reviewing photos.

General Audience Summary

People take photos to capture and revisit memories, especially with the rise of digital cameras and smartphones. Indeed, more than 1 trillion photos are taken annually worldwide. Further, access to smartphones and artificial intelligence technologies makes it easy to review photos, with features such as digital photo albums (e.g., the iPhone's Memories feature) or reminders (e.g., Facebook's memories) encouraging people to review their photos more frequently. But how do features of the photos people take, and hence later review, affect how people will remember their own experiences? We focus on one such feature: the viewpoint depicted in the photos. We distinguish between *first-person viewpoint*, where the photo shows the scene as the photographer saw it originally, versus *third-person viewpoint*, where the photo depicts a perspective not seen originally, such as photos that include the photographer themselves, generally taken by someone else or as a selfie. We argue and find that when people review photos that include themselves, memories of their own experiences shift, taking on more of an observer's memory perspective rather than that of their original experiences. Across five experiments—two in natural settings and three online using respondents' past experiences—we consistently find that reviewing third-person photos leads to this shift. Prior research either relied on correlational evidence (King et al., 2024) or prompted respondents to recall memories specifically from an observer's perspective (Butler et al., 2016) to demonstrate that memories can shift. Our work shows that merely reviewing one's own photos can naturally cause this shift without any explicit instructions to adopt an observer perspective. Our findings also suggest that reviewing a mix of first-person and third-person photos can reduce such a memory shift, such that people recall the experience more from its original perspective.

Keywords: remembering, photos, memory perspective, actor, observer

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continued

People engage in experiences in part because they want to look back at them (Tully & Meyvis, 2017), and having photos allows them to do that. With the rise of digital cameras and smartphones, people have been taking photos for a variety of reasons (Van House, 2011), to the tune of over 1 trillion a year worldwide (Broz, 2024). Importantly, the ubiquity of smartphones not only allows people to effortlessly take photos but also provides easy access when people want to look back at their experiences. Further, technologies that use artificial intelligence to create digital photo albums (e.g., the iPhone's Memories feature) also encourage people to review their photos more frequently.

One important aspect of people's memories is the perspective with which an experience is recalled. First-person versus third-person memory perspective has been linked to a wide range of important outcomes, from situational attribution (Frank & Gilovich, 1989) to assessments of self-change (Libby et al., 2005). Given the prevalence of personal photos as memory cues, we investigate how features of the photos people review naturally affect how people will remember an experience. We argue that a key feature is the viewpoint depicted by the photo, namely whether a photo depicts a first-person versus third-person viewpoint. For example, imagine being on a tour of New York City, stopping at Washington Square Park to see the famous arch. You have two photos of the experience: one captures your first-person viewpoint of the arch, and another captures a third-person viewpoint of you in front of the Arch (see Figure 1 for an illustration). While taking (vs. not taking) different types of photos may affect memory through encoding (Barasch et al., 2017), we are specifically interested in the role of photos when retrieving those memories and ask: Could reviewing the two types of

photos depicted in Figure 1 differentially change how you remember your visit?

We suggest that reviewing photos taken from a third-person (vs. first-person) viewpoint may shift people's memories away from their original perspective, that is, a form of memory distortion (Siedlecki & Falzarano, 2024). We argue that even though people reviewing their photos had the original experience (i.e., they were the actors in this experience), reviewing photos taken from a third-person viewpoint will shift memories away from their original actor memory perspective toward that of an observer. We study this effect across five experiments: two conducted in the field, where a meaningful experience was created in a natural setting, and three online studies that relied on respondents' past experiences. We find causal evidence that reviewing photos depicting a third-person viewpoint shifts memories toward an observer's memory perspective.

Reviewing Photos and Memory

When recalling one's experiences, people generate mental images that reveal a particular perspective (Nigro & Neisser, 1983). People can recall their experiences from an actor's memory perspective (i.e., as seen through their own eyes) or from an observer's memory perspective (i.e., how a bystander would see it; St. Jacques, 2019), sometimes holding both perspectives for the same experience (Rice & Rubin, 2009). As expected, autobiographical memories are recalled mostly as they were originally experienced: from an actor's memory perspective (Marcotti & St Jacques, 2018). However, memory perspective can change over time: older memories are more likely to be recalled from an observer's memory perspective than more recent

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contributed to validation: verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs. Kristin Diehl and Minjeong Ko contributed to visualization: preparation, creation, and/or presentation of the published work, specifically visualization/data presentation. Kristin Diehl, Alixandra Barasch, Minjeong Ko, and Gal Zauberman contributed to writing—original draft: preparation, creation, and/or presentation of the published work, specifically writing the initial draft (including substantive translation). Kristin Diehl, Alixandra Barasch, Minjeong Ko, and Gal Zauberman contributed to writing—review and editing: preparation, creation, and/or presentation of the published work by those from the original research group, specifically critical review, commentary, or revision: including pre- or postpublication stages.

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 The preregistered design and analysis plan is accessible at https://aspredicted.org/ZT8_67W (Study 3), https://aspredicted.org/MQ7_125 (Study 4), and https://aspredicted.org/H7N_L7F (Study 5).

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Figure 1

Photo Taken From a First-Person Viewpoint (Left) Versus a Third-Person Viewpoint (Right)



Note. Photo on the left is from “Spending Christmas in New York: Festive Things to Do in NYC,” by Gratsiela Georgieva, December 19, 2020, *Blushrougette* (<https://blushrougette.com/travel/us-destinations/spending-christmas-in-new-york/>). Copyright 2020 by Sam Palmen. Reprinted with permission. Photo on the right is from *Photos for Washington Square Arch*, by Shannon T., December 28, 2020, *Yelp* (https://www.yelp.com/biz_photos/washington-square-arch-new-york?select=sEGLd0aPeO2rOXXpKDu2JA). Copyright 2020 by Shannon T. Reprinted with permission. See the online article for the color version of this figure.

memories (e.g., Rice & Rubin, 2009; Robinson & Swanson, 1993). Prior research explicitly instructed respondents to recall events from either perspective to assess consequences (Butler et al., 2016; Marcotti & St Jacques, 2018; Robinson & Swanson, 1993). Building on this past work, we are interested in the antecedents that cause people to adopt an observer’s memory perspective outside the lab (and outside the passage of time).

Investigating the role of photos on memory, King et al. (2024) found that when people reported having a photo of a particular autobiographical event that included themselves, they deemed the event more important, had reviewed the photo more often, and remembered the experience more from an observer’s perspective. While suggestive, the correlational nature of these findings does not allow pinpointing the role photos play in how people remember an experience. For example, it is unclear whether photos operate at encoding and/or retrieval and what effect reviewing specific types of photos (vs. recalling the event *per se*) has. In our article, we are specifically interested in the causal effect of reviewing photos on memory perspective and in distinguishing it from encoding. We suggest and show causally that reviewing one’s own photos taken from a third-person viewpoint can shift memories toward an observer’s memory perspective. This shift occurs naturally without prompting and persists over time.

As mental representations of events, first-person versus third-person perspectives serve a critical function in many psychological processes with important consequences for judgments, emotions, and behavior (Libby & Eibach, 2011). In the context of reviewing photos, consequences for the memories themselves (King et al., 2024) are particularly relevant. Without reviewing photos, recalling an

experience from an observer’s memory perspective can lead to less vivid memories, compared to recalling the experience from an actor’s perspective. (Butler et al., 2016; Marcotti & St Jacques, 2018; Nigro & Neisser, 1983; Rice & Rubin, 2009). If reviewing a third-person viewpoint shifts memory perspective toward more of an observer perspective, this prior research suggests that memory vividness may suffer. However, even if reviewing photos shifts the memory perspective in that direction, the effect on vividness may not be the same, because photos themselves may also increase memory vividness. Indeed, King et al. (2024) found evidence that the availability of photos was associated with more vivid memories; however, the authors did not find any differences in vividness as a function of the viewpoint depicted in the photo. We examine the causal effect of reviewing different types of photos on memory vividness in Studies 2 and 5.

Note that recalling events can serve many functions. It may thus be beneficial to adopt different memory perspectives depending on what goal is activated (Libby, 2024; Niese et al., 2024). We will discuss the implications of our work for this line of reasoning in the General Discussion.

Transparency and Openness

We describe our sampling plan, all data exclusions (if any), all manipulations, and all measures in the studies or the [Web Appendix, Supplemental Material](#). The design and analyses for Studies 3, 4, and 5 were preregistered. All data and analysis code for Studies 1–5 are available at https://researchbox.org/2598&PEER_REVIEW_passcode=DHJZRL. Data were analyzed using SPSS 28.

All measures taken in each study are reported in the [Web Appendix, Supplemental Material](#).

Study 1: Can Reviewing Photos of a Real-Life Event Alter Memory Perspective?

In this field study, we recruited individuals visiting Washington Square Park in New York City. We either took a Polaroid of them in front of the Arch (*third-person viewpoint*) or took a Polaroid for them of the Arch (*first-person viewpoint*). They were given these photos to keep, thus manipulating the type of photo available to review. Several weeks later, we contacted participants for a follow-up study, assessing memory perspective. We predict that those who possess and review a photo of themselves in front of the arch remember the experience more from an observer's memory perspective than those who possess and review a photo of just the arch.

Method

Participants

The first phase of this study was completed by 255 people (62.35% female, 36.08% male, 1.57% missing response; $M_{\text{age}} = 28.20$). Two months after their visit to Washington Square Park, all participants were sent an email with a postexperience survey they could complete for \$3. Of those participants who successfully completed the survey at the park, 126 (65.1% female; 33.3% male, 1.6% did not answer, $M_{\text{age}} = 29.02$) completed the postexperience survey. Importantly, there was no significant difference in response rate between conditions; *first-person viewpoint*: 47.5% and *third-person viewpoint*: 51.1%, $\chi^2(1) = .17$, $p = .68$.

Procedure

Research assistants (RAs) approached individuals who were near the famous Washington Square Arch in New York City. RAs asked whether they would be willing to take a Polaroid photo and complete a short survey about their experience in exchange for taking the Polaroid photo home and having a chance to win a \$100 gift card. For groups, RAs approached only one person in the group to participate in the study. Data collection took place over 15 days

in November/December and in April due to weather conditions. Sample size was determined by participation during those days.

Participants were randomly assigned to one of two conditions (*first-person vs. third-person viewpoint*) as a function of the hour they participated (condition rotated every hour). In the *first-person viewpoint* condition, the RA took a Polaroid of the arch for the participant, while in the *third-person viewpoint* condition, the RA took a Polaroid of the participant standing in front of the arch (similar to the examples in [Figure 1](#)). Note, RAs took all photos for the participants, assuring that any effects can be attributed to reviewing photos and are not due to the act of taking the photo (cf. [Barasch et al., 2017](#); [King et al., 2024](#)).

While the Polaroid was being developed, participants completed a short survey with questions about their experience and provided some demographics (all measures taken in this and all other studies are fully reported in the [Web Appendix, Supplemental Material](#)).

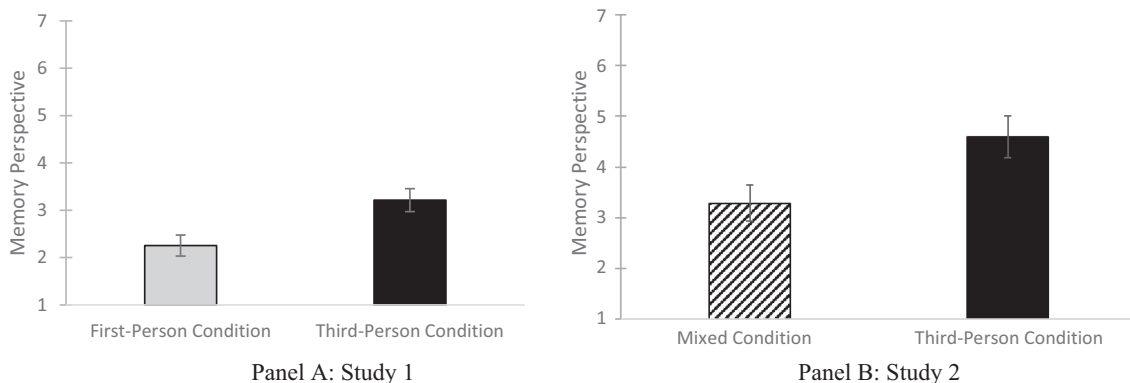
In the postexperience survey, participants were shown an image of their Polaroid photo (that had been taken by the RA originally) and were instructed to answer a few questions about their experience at Washington Square Park. First, participants were asked to think back to their Washington Square Park experience. They were told to take a moment to close their eyes and form a clear picture of the experience in their mind and to spend 15 s imagining and reflecting on the experience ([Pronin & Ross, 2006](#)). Then, participants responded to our primary measure assessing memory perspective, where they rated the image in their head on a 7-point Likert scale ranging from *mostly a first-person, actor perspective* (1) to *mostly a third-person, observer perspective* (7), with the midpoint (4) labeled as a mixture of both perspectives.

Results

One person did not respond to the memory perspective measure, so this particular analysis is based on 125 respondents. Participants who reviewed a photo that included them (*third-person viewpoint* condition) were more likely to rate their memory as being from an observer's memory perspective ($M = 3.21$, $SD = 2.03$) than those who reviewed a photo of the arch; *first-person viewpoint* condition $M = 2.25$, $SD = 1.65$, $F(1, 123) = 8.11$, $p = .005$, $\eta_p^2 = .06$. See [Figure 2](#) Panel A.

Figure 2

Effect of Reviewing Third-Person Viewpoint Photos Versus First-Person Viewpoint Photos in Study 1 (Panel A) and Versus a Mixture of Photos in Study 2 (Panel B)



Note. Higher values indicate more of an observer perspective.

Discussion

In this natural setting, we found that the type of photo people review can influence memory perspective. Reviewing photos that include the self made people more likely to remember their experience from an observer's memory perspective. This shift occurred naturally, simply as a function of reviewing photos depicting different viewpoints. Also, because RAs took these photos, these findings clearly point to differences due to retrieval rather than encoding. We conducted another study in the field to replicate these findings in a different setting and with a different comparison group. Also, in Study 1, participants reviewed the photo and responded to the memory measures within the same session. This raises the question whether memory perspective measures merely reflected the recently seen photo. To address this concern, in Study 2, participants reviewed the photo 1 week prior to the focal measures; in subsequent studies, the delay varies between 1 and 7 days.

Study 2: The Effect of Reviewing Different Types of Photos on Memory Perspective of a Real-Life Event

People not only like photos of themselves but also take and review photos that depict their first-person perspective. In this study, we compare the effect of reviewing only a photo that features oneself (i.e., *only* a third-person viewpoint) with reviewing such a photo in combination with a photo that depicts one's own first-person viewpoint. We predict that people reviewing only a photo featuring themselves will remember their experience more from an observer's memory perspective compared to those reviewing two photos that depict both a first-person and a third-person viewpoint.

Method

Participants

Participants were recruited on the campus of the University of Southern California over three separate days ($N = 170$; 44.1% female, 55.9% male, $M_{\text{age}} = 24.12$). Sample size was determined by participation across these 3 days. Participants received their picture via email, and for every 25 participants, we raffled off a \$25 gift card. Participants who agreed to be recontacted ($N = 102$) received a survey link 7 months later and chose whether to respond ($N = 46$, 45.7% female, 54.3% male, $M_{\text{age}} = 23.32$).

Procedure

We asked potential participants if they were interested in taking a photo of themselves by sticking their head through a cardboard cutout of Tommy Trojan, one of the university's symbols. After having their photo taken, participants completed a short computer-based survey about their experience and indicated whether we could contact them again. After 10 days, participants received either only the photo of themselves (*third-person viewpoint* condition) or, in the *mixed viewpoint* condition, the photo of themselves and a picture of what they had seen while taking the picture (i.e., the photographer and scenery behind the photographer; see Figure 3).

Those responding to the follow-up survey 7 months later responded to the same memory perspective measure as in Study 1. Next, they were shown the photo(s) they had received originally and responded to a few additional measures. Note, participants could still have had access to the photos they had received originally. We did not specify that participants should not review the photos prior to the last survey, so as not to trigger any demand

Figure 3

Participants in the Third-Person Viewpoint Condition Received Only Their Own Photo (Left), Participants in the Mixed Viewpoint Condition Received Both Types of Photos



Note. See the online article for the color version of this figure.

effects. A week later, participants were contacted again ($N = 36$, 47.2% female, 52.8% male, $M_{\text{age}} = 23.66$). We did not find any significant differences in response rate between conditions; *third-person*: 25.9% and *mixed*: 16.5%, $\chi^2(1) = 1.78$, $p = .18$. They responded to the same focal memory perspective measure as before. We also assessed how vividly participants remembered the experience on a scale from 1 (*not at all vividly*) to 15 (*extremely vividly*).

Results

A one-way analysis of variance (ANOVA) revealed that those who reviewed only a photo of themselves (*third-person viewpoint* condition) remembered the experience more from an observer's memory perspective ($M = 4.59$, $SD = 1.92$) compared to those in the *mixed viewpoint* condition, $M = 3.29$, $SD = 1.33$, $F(1, 34) = 4.95$, $p = .033$, $\eta_p^2 = .13$; see Figure 2 Panel B. However, despite this difference in memory perspective, memory vividness did not differ between conditions, $M_{\text{Third Person}} = 9.18$, $SD = 3.14$; $M_{\text{Mixed}} = 10.07$, $SD = 2.76$; $F(1, 34) = .75$, $p > .39$.

For robustness, we also estimated a one-way analysis of covariance (ANCOVA), controlling for participants' memory perspective prior to reviewing their photo(s). Memory perspective assessed before reviewing the photos was significantly related to memory perspective a week after reviewing the photos, $b = .46$, $t(34) = 3.56$, $p = .001$. Controlling for this baseline memory perspective, the effect of photo viewpoint persisted, $F(1, 33) = 4.15$, $p = .05$, $\eta_p^2 = .11$. Those who reviewed only a third-person viewpoint photo (least square mean = 4.49, $SE = .321$) remembered the experience directionally more from an observer's memory perspective than those who reviewed both first-person and third-person viewpoint photos (least square mean = 3.44, $SE = .404$).

Discussion

This study finds that people who only review photos depicting themselves remember their experience more from an observer's memory perspective, compared to those who review such a photo along with one depicting a first-person viewpoint. Again, because photos were taken by RAs, these findings can be causally identified as reviewing rather than encoding effects. Notably, the effect persisted for a week between manipulated reviewing and recalling the experience. We also assessed but did not observe any statistically significant effect of reviewing different types of photos on reported memory vividness.

Combined, the first two studies suggest that reviewing different types of photos can naturally alter people's memory perspective. Given that these studies were conducted in the field and that the delay between experience and subsequent surveys was substantial, our sample sizes were relatively small. Hence, we conducted three additional, preregistered studies with a larger set of respondents to ensure robustness and generalization to other settings.

Study 3: The Effect of First-Person Versus Third-Person Viewpoint Photos on Memory Perspective of a Personal Experience

In Studies 1 and 2, all participants had the same real-life experience, and access to photo type was manipulated. In Studies 3–5,

participants were screened to have both first-person and third-person viewpoint photos from recent experiences of their choosing. We manipulated whether they subsequently reviewed a third-person or first-person viewpoint photo. In this study, we also introduced a *control* condition, where participants uploaded but did not further review their photos, to more precisely examine the impact of reviewing one's photos.

Method

Participants

At Time 1, 199 undergraduate students from a private U.S. university (46.7% female, 53.3% male, $M_{\text{age}} = 19.77$) participated for partial course credit by uploading two photos. Sample size was determined by subject pool allocation (max = 200) and participation. Out of those, 164 also participated at Time 2 for additional course credit (44.5% female, 55.5% male, $M_{\text{age}} = 19.77$) and responded to the focal measure. There were no significant differences in response rate between conditions; *third-person*: 82.8%, *first-person*: 86.6%, and *control*: 77.9%, $\chi^2(2) = 0.31$, $p = .86$. As preregistered (https://aspredicted.org/ZT8_67W), we analyzed all responses provided at Time 2. When we exclude 13 participants who did not follow instructions regarding the photo upload manipulation (e.g., uploaded the same photo twice, etc.), the pattern of results remains the same (see Web Appendix, Supplemental Material).

Procedure

At Time 1, participants first briefly described an event they had experienced within the last 4 weeks, where they had photos that both included and did not include themselves. Respondents described a wide range of events, such as attending sporting events, engaging in outdoor activities, traveling, etc. (see Web Appendix, Supplemental Material for types of participant experiences). They were then randomly assigned to one of three conditions (*first-person viewpoint*, *third-person viewpoint*, *control*) in a between-subjects design. Depending on their assigned condition, participants uploaded two photos that did not include themselves (*first-person viewpoint* condition) or two photos that included themselves (*third-person viewpoint* condition). For first-person viewpoint photos, participants also indicated whether they recognized any people featured in the photos. For third-person viewpoint photos, participants indicated whether each photo was a selfie. Subsequently, we asked participants in these two experimental conditions to review the two photos they had uploaded four times, each time rating the photo on a different aesthetic dimension (colorful, detailed, blurry, dark). Providing these ratings constituted our manipulation of reviewing different photo types. Evaluations of uploaded photos did not differ on these aesthetic dimensions. In the *control* condition, participants uploaded one photo of each type, ensuring they possessed both types of photos. Similar to the experimental conditions, they indicated whether they recognized any people in the photos (first-person viewpoint photo) and whether the photo was a selfie (third-person viewpoint). However, in the control condition, participants did not rate these photos at T1; thus, their review was limited to selecting and uploading the photos. To equalize time spent in the study, participants in the *control* condition evaluated the brand personality of two brands (i.e., Nike and Adidas).

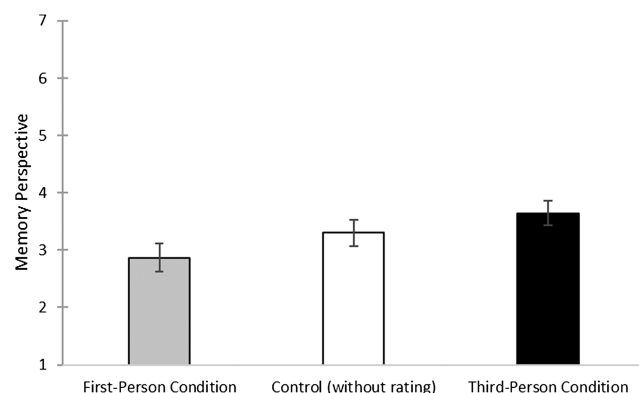
Participants were to receive an invitation to participate in Part 2 of the study 1 day after they completed Part 1. However, for both technical reasons (i.e., some participants received the invitation the same day they completed Part 1) and compliance reasons (i.e., participants did not complete the survey on the day they received the invitation), participants took between 0 and 6 days to respond to the second part of the study (*Mode and Mdn* = 1, *M* = 1.47, *SD* = 1.14). Average days passed between Time 1 and Time 2 did not differ significantly between conditions (*ps* > .44). At Time 2, we assessed participants' memory perspective using the same measure as in previous studies. For symmetry, after completing the memory perspective measure at Time 2, those in the control group also evaluated their photos on the same four aesthetic dimensions as those in the experimental conditions did at Time 1 (see [Web Appendix, Supplemental Material](#) for detail).

Results

To test our prediction that reviewing photos that feature the self leads people to remember their experience more from an observer's memory perspective, as preregistered, we estimated planned contrasts (see [Figure 4](#)). Planned contrasts indicated that people who reviewed photos that included themselves were more likely to remember their experience from an observer's memory perspective (*M* = 3.64, *SD* = 1.63) than those who reviewed photos that did not include themselves, *M* = 2.87, *SD* = 1.74, $F(1, 161) = 5.74, p = .02, \eta_p^2 = .03$. Memory perspective for the *control* condition fell in between the two experimental conditions (*M* = 3.30, *SD* = 1.71). We did not observe any significant difference between the *control* condition and either of the experimental conditions (vs. *first-person viewpoint*: $p > .18, \eta_p^2 = .01$; vs. *third-person viewpoint*: $p > .29, \eta_p^2 = .01$). For robustness, we also conducted a repeated-measures ANOVA with memory perspective assessed at Time 1 and Time 2, and an ANCOVA controlling for the time lag between reviewing the photos at Time 1 and responding to the focal measure at Time 2. Results mirror those of the univariate ANOVA analysis and are reported in the [Web Appendix, Supplemental Material](#).

Figure 4

Effect of Reviewing Photos Depicting Different Viewpoints Versus Control (Without Rating) on Memory Perspective



Note. Higher values indicate more of an observer perspective.

Discussion

These results replicate our findings from the field: compared to reviewing photos that are taken from a first-person viewpoint, reviewing photos that depict a third-person viewpoint leads to recalling the experience more from an observer's memory perspective. This was the case for participants' own photos across a wide range of experiences. Note, however, that here and in subsequent studies, across conditions, memories were still held mostly from an actor's perspective. This is likely due to the relatively recent experiences (i.e., within the last month) these lab experiments focused on ([Rice & Rubin, 2009](#)).

Study 4: The Effect of Reviewing Only Third-Person Viewpoint Photos Versus Reviewing a Mixture of Photo Types on Memory Perspective of a Personal Experience

In real life, people have a variety of pictures to review, some depicting a first-person viewpoint and some a third-person viewpoint. As in Study 2, we examine the effects of reviewing only third-person viewpoint photos, compared to reviewing photos that depict both first-person and third-person viewpoints. In line with previous studies, we predict that reviewing only photos featuring oneself leads people to remember their experience more from an observer's memory perspective compared to reviewing photos that include a mixture of perspectives.

Method

Participants

At Time 1, 199 undergraduate students from a private U.S. university (40.7% female, 59.3% male, *M*_{age} = 19.8) participated for partial course credit. Sample size was determined by subject pool allocation (max = 200) and participation. Out of those, 173 also participated at Time 2 for additional course credit (43.4% female, 56.6% male, *M*_{age} = 19.7) and responded to the focal measure. We did not find any significant differences in response rate between conditions; *third-person viewpoint*: 90.1%, *mixed viewpoint*: 83.7%, $\chi^2(1) = 0.24, p = .63$. We analyzed all responses collected at Time 2. When we exclude seven participants who did not provide meaningful responses in the event description section or uploaded the same photo twice, as we had preregistered (https://aspredicted.org/MQ7_125), the pattern of results is the same (see [Web Appendix, Supplemental Material](#)). Similar to Study 3, for both technical and compliance reasons, days elapsed between participation in Part 1 and Part 2 ranged from 0 to 4 days (*Mode and Mdn* = 1, *M* = 0.89, *SD* = 0.73), but did not differ significantly between conditions ($p > .30$).

Procedure

Participants were randomly assigned to one of two conditions (*third-person viewpoint*, *mixed viewpoint*). As in Study 3, participants in the *third-person viewpoint* condition uploaded two photos that both included themselves. Those in the *mixed viewpoint* condition uploaded one photo that included themselves and one that did not. Note though that while participants in the *mixed viewpoint* condition uploaded similar photos as those in the *control* condition in Study 3, in this study, all participants (in both conditions) rated the photos they uploaded four times on aesthetic dimensions. Thus,

both conditions reviewed the photos similarly. Further, in Study 2, participants in the *third-person viewpoint* condition only reviewed a single third-person viewpoint photo compared to two photos in the *mixed viewpoint* condition. In this study, we held the number of photos reviewed constant across conditions. As in Study 3, the focal measure of memory perspective was asked at Times 1 and 2 along with other measures. See [Web Appendix, Supplemental Material](#) for a depiction of the flow of the experiment.

Results

As expected, people exposed only to photos that featured themselves (*third-person viewpoint* condition) remembered their experience more from an observer's memory perspective ($M = 3.54$, $SD = 1.85$) than those exposed to a mixture of photos, $M = 2.96$, $SD = 1.65$; $F(1, 171) = 4.61$, $p = .03$, $\eta_p^2 = .03$. For robustness, as in Study 3, we also conducted a repeated-measures ANOVA and an ANCOVA controlling for the time lag between Time 1 and Time 2. Results mirror those of the univariate ANOVA analysis and are reported in the [Web Appendix, Supplemental Material](#).

Discussion

Study 4 replicates prior findings, showing that reviewing photos exclusively depicting a third-person viewpoint leads to memories characterized as more from an observer's memory perspective than reviewing a mixture of photos. In Study 5, we include a *control* condition to assess whether a mixture of photo types preserves memory perspective more or less compared to control.

Study 5: Further Evidence of the Effects of Reviewing Only Third-Person Perspective Photos

In this study, we also included a *control* condition where people simply uploaded but did not repeatedly rate their photos (similar to Study 3). We predict that repeatedly reviewing only photos featuring oneself (*third-person viewpoint* condition) leads people to remember their experience more from an observer's (vs. actor's) memory perspective compared to a mixture of photos (*mixed viewpoint* condition) and compared to control.

Method

Participants

At Time 1, 360 undergraduate students from a private U.S. university (46.4% female, 41.9% male, 11.7% N/A, $M_{\text{age}} = 19.7$) participated for partial course credit. Sample size was determined by subject pool allocation (max = 400) and participation. Out of those, 211 also participated at Time 2 for additional course credit (48.8% female, 43.1% male, $M_{\text{age}} = 19.5$) and responded to the focal measure. There were no significant differences in response rate between conditions; *third-person viewpoint*: 58.7%, *mixed viewpoint*: 60.7%, *control*: 56.6%, $\chi^2(2) = 0.17$, $p > .9$. We analyzed all responses obtained at Time 2. Excluding nine participants, who did not provide meaningful responses in the event description section or uploaded the same photo twice, as preregistered (https://aspredicted.org/H7N_L7F), did not change the results (see [Web Appendix, Supplemental Material](#)). As in Study 3, for both technical and

compliance reasons, time elapsed between participation in Part 1 and Part 2 ranged from 0 to 7 days ($Mode \text{ and } Mdn = 1$, $M = 1.94$, $SD = 1.70$), but did not differ significantly between conditions ($ps > .5$).

Procedure

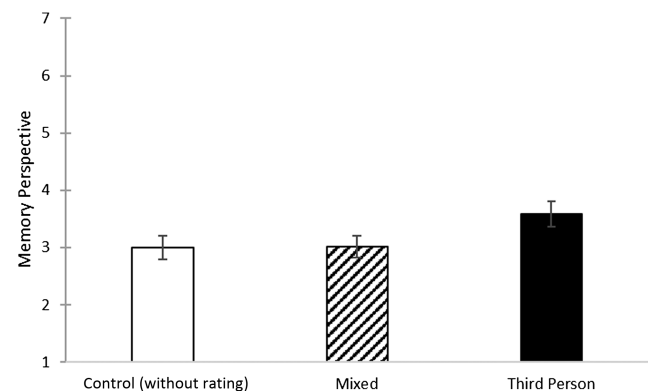
Participants were randomly assigned to one of three conditions (*third-person viewpoint*, *mixed viewpoint*, *control*). As before, participants in the *third-person viewpoint* condition uploaded two photos that both included themselves. Those in the *mixed viewpoint* and the *control* conditions uploaded one photo that included themselves and one that did not. While participants in the *mixed viewpoint* and the *control* conditions uploaded the same types of photos, the procedures after uploading these photos differed. Participants in the *mixed viewpoint* condition rated their uploaded photos four times, each time on different aesthetic dimensions (just like the *third-person viewpoint* condition), thus increasing the opportunity to actively review their photos. However, as in Study 3, those in the *control* condition did not rate their photos and thus did not further review their photos after they were uploaded. Also, as in Study 3, the focal measure of memory perspective was asked at Times 1 and 2 along with other measures. See [Web Appendix, Supplemental Material](#) for a depiction of the flow of the experiment. Assessing vividness of visual memory, at Time 2, participants responded to the prompt "While remembering the event, I can see it in my mind" on 7-point scale labeled 1 (*not at all*), 3 (*vaguely*), 5 (*distinctly*), and 7 (*as clearly as if it were happening now*), adopted from [Berntsen et al. \(2019\)](#).

Results

As predicted, people who reviewed only photos that featured themselves (*third-person viewpoint* condition) remembered their experience more from an observer's memory perspective ($M = 3.58$, $SD = 1.87$) than those who reviewed a mixture of photos, $M = 3.00$, $SD = 1.77$; $F(1, 208) = 3.85$, $p = .05$, $\eta_p^2 = .018$. This was also the case compared to the *control* condition, where they did not actively review any photos, though this effect is marginal, $M = 3.01$, $SD = 1.61$; $F(1, 208) = 3.60$, $p = .06$, $\eta_p^2 = .017$; see [Figure 5](#). There was

Figure 5

Effect of Reviewing Photos of Different Viewpoints Versus Control (Without Rating) on Memory Perspective



Note. Higher values indicate more of an observer perspective.

no significant difference between the *mixed viewpoint* and *control* conditions ($p > .96$), suggesting actively reviewing a mixture of photos did not shift memory perspective compared to simply uploading such photos. For robustness, we also conducted a repeated-measures ANOVA and an ANCOVA controlling for the time lag between Time 1 and Time 2. Results mirror those of the univariate ANOVA analysis and are reported in the [Web Appendix, Supplemental Material](#).

A one-way ANOVA on memory vividness revealed no significant differences between conditions ($M_{\text{Third Person}} = 5.21$, $SD = 0.96$; $M_{\text{Mixed}} = 5.32$, $SD = 0.95$; $M_{\text{Control}} = 5.17$, $SD = 1.06$, $p > .64$).

Discussion

The present study replicates prior findings, showing that exclusively reviewing photos that depict a third-person viewpoint leads to more of an observer's memory perspective. In addition, this study also shows that it is reviewing photos that *only* offer a third-person viewpoint that has this effect. Reviewing a mixture of photos repeatedly does not alter one's memory perspective compared to reviewing them only once when uploading. Further, as in Study 2, we did not observe any significant effects on vividness.

General Discussion

The last decade has seen an explosion in photo taking. People often take photos because they believe that reviewing these photos helps them remember their experiences. Indeed, research shows that photo review focuses attention on depicted aspects and their associated details (Henkel et al., 2021), providing evidence that reviewing photos shapes memory. Given the centrality of mental perspective in the study of memories (Libby & Eibach, 2011), we examine the causal link between reviewing different types of photos and people's memory perspective. We find that reviewing photos depicting a first-person viewpoint leads people to remember their experience more from their original, actor perspective, whereas reviewing third-person photos that include the self shifts memories toward an observer perspective.

Theoretical and Practical Implications

Our research contributes to work on the antecedents of memory perspectives (Nigro & Neisser, 1983). While prior research, in the absence of photos, prompted people to adopt an actor versus observer point of view when remembering events, we show that such shifts can occur spontaneously and unprompted, through the simple act of reviewing one's photos—an increasingly common behavior in everyday life. In our studies, people reviewed personal photos from a wide range of past events that had occurred 1–7 months earlier, with the effect of reviewing photos persisting for up to 1 week afterward.

Moreover, prior research that examined the effect of personal photos on memory retrieval (King et al., 2024) relied on self-reports of the existence of photos, limiting causal interpretations. That approach also could not determine whether shifts in memory perspectives stemmed from encoding at the time of the experience or from reviewing photos afterward. In contrast, our approach either created or verified the existence of photos, allowing us to isolate the

unique causal effect of reviewing photos on memory perspective. We thus contribute to research on the role of photos in autobiographical retrieval and in constructive memory processes (Fawns, 2020). We also believe that our experimental approach—manipulating photo review in a realistic, memory-unrelated task (i.e., rating aesthetic features)—adds to the methodological toolbox for investigating the causal impact of photos and perspectives in other contexts.

People frequently expose themselves to third-person viewpoint photos in daily life (e.g., displaying them on refrigerators), which our findings suggest can influence how experiences are later remembered. For those wishing to preserve their original memory perspective, our results suggest reviewing a mix of first- and third-person photos is more effective than relying solely on third-person images. That said, people may seek out different types of photos for different goals: first-person photos to remember the actual experience and third-person photos to reflect on its meaning (Niese et al., 2024). Hence, access to varied perspectives (as in our mixed conditions) may not only reduce memory perspective shifts but may also facilitate a wider range of memory functions (e.g., recalling concrete details vs. abstract meanings). Smartphones with dual-facing cameras and apps like “BeReal” allow and compel people to create a varied photo stock. Artificial intelligence-curated photo albums (e.g., iPhone Memories) may benefit from purposefully incorporating a mixture of photo types.

Further, access to different types of photos may promote flexibility in adopting different memory perspectives. In fact, Libby (2024) proposes that such *flexibility* could lead to richer and potentially more accurate memories (see also Schacter & Addis, 2007). Whether this is the case remains an open question for future research.

Limitations and Future Research

Prior research has found that third-person perspectives can reduce memory vividness (e.g., Butler et al., 2016). We did not observe such effects, possibly because reviewing personal photos already heightened vividness. This raises the possibility that perspective shifts from reviewing own photos may, at times, lead to different outcomes than shifts from mental recall alone. Future research could explore consequences uniquely tied to photo review, such as its impact on perceived personal change (Libby et al., 2005).

In our lab studies, people reviewed the same small set of photos repeatedly over a few minutes. In everyday life, people might review a larger set of photos once, but for longer (e.g., browsing a photo album), or view the same photo briefly but repeatedly (e.g., one displayed on a fridge). Notably, Study 5 found no difference between reviewing a mixed set of photos once during uploading and reviewing it repeatedly afterward. Future research could investigate how the intensity and timing of photo review influence memory perspective.

Furthermore, in our research, reviewing third-person photos shifted memories toward an observer perspective, though memories overall remained anchored in the actor perspective. This may reflect the relative recency of the experiences examined, as older memories are more likely to be recalled from an observer perspective (e.g., Rice & Rubin, 2009). Future research could examine whether third-person photos have stronger effects on older memories.

In sum, given the growing prevalence of photos in daily life and the rise of artificial intelligence-supported tools that encourage photo review, these and other open questions merit deeper investigation.

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